SynthWake: LLM-Based Speech Synthesis for Data-Efficient Wake-Word Verification

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Abstract

Deploying new wake words for speaker verification typically requires collecting large-scale text-dependent recordings of the target phrase, which is costly and hard to scale. We propose **SynthWake**, which uses LLM-based speech synthesis to convert text-independent (TI) utterances into synthetic text-dependent wake-word data. Given labeled TI speech, we synthesize target wake-word utterances using zero-shot text-to-speech (TTS) models, with speaker utterances serving as prompt audio, eliminating the need to collect real recordings of newly introduced wake-word phrases. To bridge the synthetic–real gap, we introduce domain adversarial training and a speaker-aware cross-domain center alignment loss to align speaker representations across domains. On the SMIIP-TV wake-word benchmark, SynthWake achieves 3.938% EER and a 14.7% relative FRR reduction at FAR = 1%, outperforming the synthetic-data baselines and performing competitively with the HI-MIA-trained real-TD reference under a cross-corpus comparison. These results demonstrate that synthetic wake-word speech combined with speaker-aware alignment provides an effective alternative to collecting large-scale phrase-matched real recordings for each newly deployed target phrase.

Keywords: Wake word verification, speech synthesis, data efficiency, large language model

1. Introduction

Speaker verification (SV) is essential for authenticating users on always-listening devices, such as smart speakers and in-vehicle assistants. Since interactions on these devices typically begin with a predefined wake word, text-dependent speaker verification (TD-SV) is the preferred paradigm. By leveraging this fixed lexical content, TD-SV exploits both speaker identity and phrase consistency to achieve robust authentication.

In practice, state-of-the-art wake-word systems are rarely trained from scratch. A standard paradigm is to pre-train a speaker encoder on large-scale text-independent (TI) speech, followed by fine-tuning on real text-dependent (TD) recordings of the target phrase [1, 2]. While this effectively combines the broad discriminative power of TI corpora with phrase-specific supervision,

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it introduces a severe data bottleneck. Deploying any new or customized wake word requires collecting large-scale, phrase-matched recordings across diverse speakers [3]. Consequently, scaling wake-word verification to new phrases remains a costly and time-consuming process.

This work studies data-efficient wake-word speaker verification in this practical sense: we aim to reduce the need for real recordings of the target wake phrase, while still making use of existing large-scale TI speaker data. Synthetic speech is a natural way to reduce the dependence on newly collected wake-word recordings. Previous studies have used speech synthesis or voice conversion to expand limited TD-SV training sets. SynAug synthesizes text-controlled utterances to enlarge the number of speakers and phrases for TD-SV [3], while VC-AUG uses voice conversion to generate additional text-dependent samples under limited-data conditions [4]. Related TTS-based augmentation has also been explored for cross-domain speaker recognition and fixed-phrase speaker verification [5, 6]. These studies show that generated speech can provide useful supervision for speaker verification. However, they still assume a setting where task-specific real TD data, target-speaker examples, or controllable target-phrase construction is available. When no real recordings of the new wake word can be collected, conventional synthesis or voice-conversion augmentation is difficult to apply directly, because the training pipeline still needs reliable links between synthetic phrase data and real speaker distributions.

Recent zero-shot and LLM-based text-to-speech (TTS) systems change this situation. Neural codec language models and flow-matching TTS systems can synthesize speech in the voice of an unseen speaker from a short acoustic prompt and arbitrary input text. Representative systems include VALL-E and VALL-E 2 [7, 8], Voicebox [9], CosyVoice [10], F5-TTS [11], MaskGCT [12], and IndexTTS [13]. This capability suggests a new route for wake-word verification: instead of collecting real recordings of a new wake word, one can use existing labeled TI utterances as speaker prompts and synthesize the target wake word for those same speakers. In this way, large-scale TI speech can be converted into phrase-relevant TD training data.

However, zero-shot TTS does not make synthetic speech a drop-in replacement for real speaker data. Although modern TTS models can preserve speaker similarity perceptually, generated speech can still differ from real recordings in prosody, duration, channel characteristics, codec artifacts, and speaker-embedding distribution. Recent work on zero-shot TTS for short-utterance speaker verification reports that combining real and synthetic speech can help, but that synthetic speech does not provide the same benefit as real speech when used as a replacement [14]. Our experiments show a similar phenomenon at training scale: synthesized wake-word speech is useful when paired with real TI utterances from the same speakers, but removing these real speaker anchors can erase the gains and even increase the EER above the baseline. This finding motivates a synthetic-to-real adaptation strategy rather than a naive data-mixing strategy.

We propose **SynthWake**, a data-efficient framework for wake-word verification designed to mitigate the aforementioned identity drift and synthesis artifacts. Given labeled TI speech, SynthWake uses IndexTTS to synthesize the target wake word for the same speakers, constructing a synthetic TD corpus without collecting real recordings of the new phrase. To make the synthetic data useful for a strong SV model, we introduce two complementary alignment mechanisms to bridge the synthetic–real domain gap. First, domain-adversarial learning with a gradient reversal layer reduces global synthetic–real mismatch. Second, speaker-aware cross-domain center alignment pulls each speaker's synthetic centroid toward the corresponding real centroid, using real TI utterances as stable speaker anchors. Together, these objectives encourage the model to benefit from phrase-specific synthetic supervision while avoiding performance degradation caused by domain bias.

The contributions of this work are summarized as follows:

- We study LLM-based wake-word synthesis as a scalable augmentation strategy for data-efficient TD-SV, converting labeled TI speech into target-phrase TD training data without requiring real recordings of the new wake word.
- We empirically demonstrate that synthetic wake-word speech alone is insufficient in our setting: without paired real TI anchors from the same speakers, the resulting domain and speaker mismatch can increase the EER above the baseline.
- We propose a speaker-aware synthetic-to-real adaptation framework combining GRL-based domain adversarial training with asymmetric cross-domain center alignment, effectively safeguarding the real speaker representation space from synthesis artifacts.
- On the SMIIP-TV Mandarin wake-word benchmark, SynthWake achieves 3.938% EER and a 14.7% relative FRR reduction at FAR = 1%, performing competitively with a cross-corpus real-TD reference while substantially reducing target-phrase data collection.

2. Related Works

Zero-Shot TTS and Synthetic Speech for Speaker Verification. Recent zero-shot TTS systems can generate arbitrary lexical content from short acoustic prompts. Representative approaches include neural codec language models such as VALL-E and VALL-E 2 [7, 8], flow-based systems such as Voicebox and F5-TTS [9, 11], and multilingual systems including CosyVoice, MaskGCT, and IndexTTS [10, 12, 13]. These advances make it possible to synthesize a target wake word in the voice of a speaker represented only by TI recordings.

Synthetic speech has previously been used to augment speaker-recognition and TD-SV training data. Synth2Aug studies TTS-based augmentation for cross-domain speaker recognition [5]; unit-selection synthesis has been applied to fixed-phrase verification [6]; and SynAug and VC-AUG generate additional TD samples using speech synthesis or voice conversion [3, 4]. Zhao et al. [14] further show that zero-shot TTS can improve short-utterance speaker verification when combined with real speech, but is less effective as a replacement for real data. Unlike these studies, SynthWake uses labeled TI prompts to construct training data for a newly deployed wake word without requiring real recordings of that phrase.

Domain Adaptation and Speaker-Aware Alignment. Domain-adversarial learning uses a gradient reversal layer to suppress domain-specific information [17], while statistical approaches such as maximum mean discrepancy and correlation alignment match distributions across domains [18, 19]. Related domain adaptation methods have also been studied in speaker recognition [15, 16]. However, global distribution alignment does not explicitly preserve speaker-wise correspondence. Inspired by center loss and prototypical learning [20, 21], SynthWake aligns synthesized speaker centroids with corresponding real-speech centroids while treating real representations as stop-gradient anchors.

3. SynthWake

3.1. Problem Formulation

Let $\mathcal{D}_{\text{real}}^{\text{TI}} = \{(x_i^r, y_i)\}_{i=1}^{N_r}$ denote the TI real speech dataset, where x_i^r is an utterance and y_i is the speaker label; the superscript r denotes real and s denotes synthesized throughout. Our goal

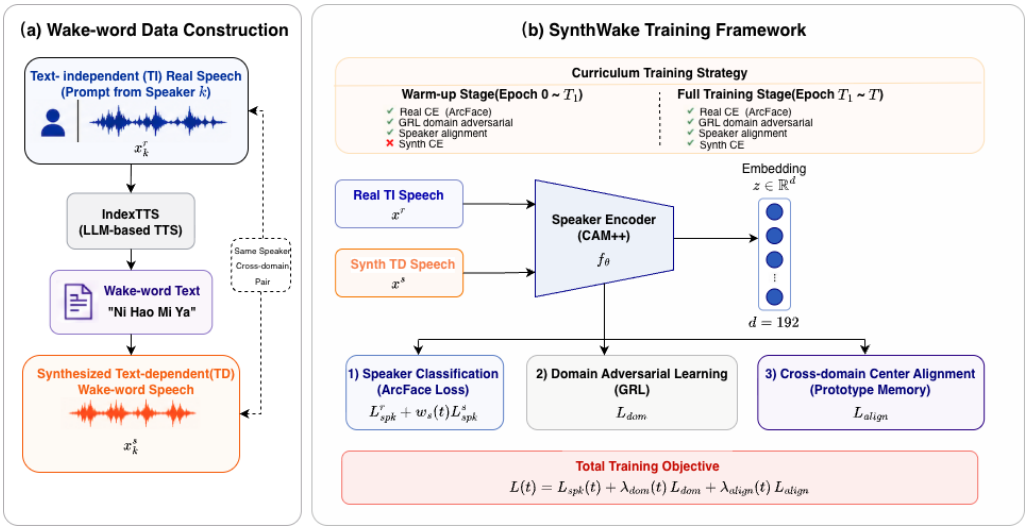


Figure 1: Overview of SynthWake. (a) Wake-word speech is synthesized from TI prompts using IndexTTS. (b) Real and synthesized speech are jointly optimized through speaker classification, domain adversarial learning, and cross-domain center alignment.

is to train a TD-SV model for a target wake word in a data-efficient manner by minimizing the need for real recordings of that wake word. We leverage IndexTTS [13] to generate wake-word audio conditioned on TI prompts. For each utterance in $\mathcal{D}^{\text{TI}}_{\text{real}}$, the audio is used as a prompt and the target wake-word text t_{wake} is synthesized as

$$x^s_j = \text{IndexTTS}(x^r_i,\, t_{\text{wake}}), \tag{1}$$

yielding a synthesized text-dependent dataset $\mathcal{D}^{\text{TD}}_{\text{synth}} = \{(x^s_j, y_j)\}^{N_s}_{j=1}$, where each synthesized utterance inherits the speaker label of its prompt.

Compared with short wake-word recordings, TI speech provides more diverse speaker-specific acoustic and prosodic cues, which can help the TTS model preserve speaker identity during wake-word synthesis. Modern LLM-based TTS systems can exploit these characteristics to generate target wake-word speech without requiring any real recording of the target phrase.

However, synthesized speech is not identical to real wake-word audio. Residual discrepancies remain in spectral characteristics, speaking style, and identity fidelity, creating a synthetic–real domain gap that can degrade verification performance. SynthWake addresses this challenge through joint representation learning and domain adaptation.

3.2. Architecture Overview

Fig. 1 illustrates the architecture. Given an input utterance x, the speaker encoder f_θ extracts an embedding $\mathbf{z} = f_\theta(x)$. Training samples come from both real and synthesized speech, and the embedding is optimized with three objectives: (1) speaker classification for discriminative identity; (2) domain-adversarial learning for global synthetic–real confusion; and (3) cross-domain center alignment for per-speaker consistency.

3.3. Domain Adversarial Representation Learning

Directly mixing real and synthetic speech may encourage the encoder to exploit domain-specific artifacts instead of speaker-discriminative cues. We therefore introduce domain-adversarial learning to reduce the global mismatch between real and synthesized embeddings.

A domain classifier $D(\cdot)$ predicts whether an embedding originates from real or synthesized speech. A Gradient Reversal Layer (GRL) is inserted between the speaker encoder and the domain classifier. During the forward pass, the GRL acts as the identity operation. During back-propagation, gradients flowing into the speaker encoder are multiplied by a negative coefficient.

The GRL coefficient follows

$$\eta(e) = \frac{2}{1 + \exp(-10p)} - 1, \qquad p = \min\left(\frac{e}{e_{\text{adv}}}, 1\right), \tag{2}$$

where e is the one-based epoch index and $e_{\text{adv}} = 2$. The domain classifier is optimized to distinguish the two domains, whereas the speaker encoder is optimized to confuse it. The domain-classification loss is defined over the adaptation subset in the training objective below.

3.4. Speaker-Aware Cross-Domain Center Alignment

Domain-adversarial learning reduces marginal distribution mismatch, but it does not explicitly ensure that real and synthesized utterances from the same speaker remain close in the embedding space. We therefore introduce a speaker-aware cross-domain centroid alignment objective.

Before computing speaker centroids, the embeddings are ℓ_2-normalized. For speaker k, the real and synthesized centroids in the current mini-batch are defined as

$$\boldsymbol{\mu}^r_{k,b} = \frac{1}{|\mathcal{S}^r_{k,b}|} \sum_{i \in \mathcal{S}^r_{k,b}} \mathbf{z}_i, \qquad \boldsymbol{\mu}^s_{k,b} = \frac{1}{|\mathcal{S}^s_{k,b}|} \sum_{j \in \mathcal{S}^s_{k,b}} \mathbf{z}_j, \tag{3}$$

where $\mathcal{S}^r_{k,b}$ and $\mathcal{S}^s_{k,b}$ denote the paired real and synthesized samples of speaker k in mini-batch b, respectively.

When both real and synthesized samples of speaker k are present in the current mini-batch, the synthesized centroid is aligned directly with the current real centroid. To provide a real-speech target when no paired real sample of speaker k is present in the current mini-batch, we maintain a real-speech prototype using an exponential moving average:

$$\boldsymbol{\mu}^r_k \leftarrow \alpha \boldsymbol{\mu}^r_k + (1 - \alpha)\boldsymbol{\mu}^r_{k,b}, \tag{4}$$

where $\alpha = 0.88$ is the EMA momentum. The prototype is updated only when paired real samples of speaker k are present in the current mini-batch.

The real alignment target is defined as

$$\widetilde{\boldsymbol{\mu}}^r_{k,b} = \begin{cases} \boldsymbol{\mu}^r_{k,b}, & \text{if } \mathcal{S}^r_{k,b} \neq \emptyset, \\ \boldsymbol{\mu}^r_k, & \text{if } \mathcal{S}^r_{k,b} = \emptyset \text{ and an EMA prototype is available.} \end{cases} \tag{5}$$

Let $\mathcal{K}_b$ denote the set of speakers for which both a synthesized centroid and a valid real target are available. The cross-domain alignment loss is

$$\mathcal{L}_{\text{align}} = \frac{\gamma}{|\mathcal{K}_b|} \sum_{k \in \mathcal{K}_b} \left\| \boldsymbol{\mu}^s_{k,b} - \text{sg}\left(\widetilde{\boldsymbol{\mu}}^r_{k,b}\right) \right\|_2^2, \tag{6}$$

where sg(·) denotes the stop-gradient operation and $\gamma = 0.4$ is the internal alignment scale. The alignment loss is set to zero when $\mathcal{K}_b$ is empty.

Gradients are propagated through the synthesized embeddings, while the real centroids and EMA prototypes are treated as fixed targets. This asymmetric design prevents synthesis artifacts from pulling the real-speech representation space away from its speaker-discriminative structure. The domain-adversarial and centroid-alignment objectives are complementary: the former reduces global real–synthetic mismatch, whereas the latter preserves speaker-wise correspondence across the two domains.

3.5. Training Objective

Let $\mathcal{B}$ denote a training mini-batch, and let $w_i(t)$ denote the speaker-classification weight assigned to sample i according to its data role and the current training epoch. The weighted speaker-classification loss is

$$\mathcal{L}_{\text{spk}}(t) = -\frac{\sum_{i\in\mathcal{B}} w_i(t) \log C(\mathbf{z}_i, y_i)}{\sum_{i\in\mathcal{B}} w_i(t)}, \tag{7}$$

where $C(\cdot)$ is the ArcFace speaker classifier.

Base-real and paired-real samples use classification weights of 1.0 and 1.2, respectively. The classification weight of paired synthesized samples is set to zero during the first epoch and to 1.0 from the second epoch onward.

The adaptation subset $\mathcal{A} \subseteq \mathcal{B}$ consists of the paired-real and paired-synthesized samples drawn into $\mathcal{B}$ by a domain-balanced sampler, since only these same-speaker real–synthesized pairs are used for domain adaptation. The domain-adversarial loss is computed only over this subset:

$$\mathcal{L}_{\text{dom}} = -\frac{1}{|\mathcal{A}|} \sum_{i\in\mathcal{A}} \log p(d_i \mid \mathbf{z}_i), \tag{8}$$

where $d_i \in \{0, 1\}$ denotes the domain label.

The final training objective is

$$\mathcal{L}(t) = \mathcal{L}_{\text{spk}}(t) + \lambda_{\text{dom}}(t)\mathcal{L}_{\text{dom}} + \lambda_{\text{align}}(t)\mathcal{L}_{\text{align}}. \tag{9}$$

During the first epoch, the coefficients are set to $\lambda_{\text{align}} = 0.04$ and $\lambda_{\text{dom}} = 0.008$. From the second epoch onward, they are increased to 0.10 and 0.02, respectively.

3.6. Inference

At test time, only the speaker encoder is retained. The domain classifier, real-speech EMA prototypes, and prototype-based alignment objective are used only during training. Verification scores are computed using cosine similarity between 192-dimensional enrollment and test embeddings under the fixed-phrase TD-SV protocol.

4. Experimental Setup

4.1. Datasets

Training Data. We train on a combination of large-scale SV corpora: CN-Celeb [22], VoxCeleb2 [23], 3D-Speaker [24], and KeSpeech [25]. Corpus-specific speaker IDs are treated as disjoint. To construct paired real–synthetic data, we additionally use a filtered Mandarin subset of

Table 1: Dataset statistics. CommonVoice-ZH refers to the filtered Mandarin subset used for synthesis.

Dataset	Speakers	Utterances	Hours
CN-Celeb	2,793	650,713	1356.05
VoxCeleb2	5,994	1,092,009	2379.56
3D-Speaker	10,240	603,103	1150.83
KeSpeech	27,237	976,054	1541.69
CommonVoice-ZH	5,430	791,803	1017.50
Synthesized TD	5,430	791,732	460.11
HI-MIA (Real TD)	260	352,759	140.02
SMIIP-TV	364	24,732	12.74

CommonVoice-ZH. CommonVoice client IDs are treated as speaker labels, and utterances containing fewer than four Chinese characters are removed because preliminary experiments showed that very short prompts resulted in unstable voice cloning. The resulting subset contains 5,430 speaker IDs and is used both as real TI training data and as the prompt source for wake-word synthesis. One wake-word sample is generated for each valid prompt using IndexTTS. After discarding 71 failed syntheses, the resulting synthetic TD set contains 791,732 utterances. In total, the training data comprise 51,694 corpus-specific speaker IDs and approximately 7,900 hours of speech.

Evaluation Data. We evaluate on the "Ni Hao Mi Ya" subset of SMIIP-TV [26], which provides enrollment and test utterances of this Mandarin wake word recorded under realistic acoustic conditions. The evaluated subset contains 364 speakers after applying our protocol filtering. HI-MIA [2] is used only to train a real-TD reference system (+ Real TD) for contextual comparison and is not used in SynthWake training. Therefore, + Real TD should be interpreted as a cross-corpus reference rather than an in-domain upper bound. Statistics are summarized in Table 1.

4.2. Implementation Details

Model & Optimization. We use CAM++ [27] to extract 192-dim embeddings from 80-dim log Mel-filterbanks (25 ms window, 10 ms shift). The ArcFace classifier [28] uses margin $m = 0.3$ and scale $s = 48$. The domain classifier is a two-layer MLP (hidden dim 256, ReLU, dropout 0.1) connected via a GRL. Models are trained for 13 epochs using SGD with momentum 0.95, weight decay 10^{-4}, and a batch size of 64. The encoder and task heads use initial learning rates of 3×10^{-5} and 2×10^{-4}, respectively. The encoder is frozen during the first epoch, after which the learning rates are decayed using a cosine schedule. Parameters are averaged over the last three epochs. Data augmentation includes MUSAN noise, RIRs, and SpecAugment.

Synthesis & Curriculum. Wake-word speech is generated using IndexTTS [13]. Synthesis failures, identified by silence ratios above 40% or estimated SNRs below 10 dB, were rare, and 71 utterances were discarded. Domain-balanced sampling assigns approximately 13% of each mini-batch to paired real samples and 13% to paired synthesized samples. Base-real and paired-real samples use speaker-classification weights of 1.0 and 1.2, respectively. During the first epoch, the classification weight of paired synthesized samples is zero, while λ_{align} and λ_{dom} are set to 0.04 and 0.008. From the second epoch onward, the synthesized-sample classification weight is

Table 2: Speaker Verification Results on the "Ni Hao Mi Ya" part of SMIIP-TV. "TI": text-independent, "TD": text-dependent. Mean ± standard deviation over three runs. *Improvement over Baseline is statistically significant (Welch's t-test, $p = 0.001 < 0.01$).

Method	EER (%)	minDCF
Baseline (base corpora set only)	4.169 ± 0.03	0.387
+ Real TI data (CommonVoice-ZH)	4.132 ± 0.04	0.371
+ Synth TD (no domain adaptation)	4.007 ± 0.03	0.360
+ Real TD data (HI-MIA)	3.998 ± 0.04	0.380
SynthWake (Ours)	**3.938 ± 0.02***	**0.349**
w/o Paired Real TI Anchors	4.207 ± 0.03	0.371

set to 1.0, and λ_{align} and λ_{dom} are increased to 0.10 and 0.02, respectively. The internal centroid-alignment scale is $\gamma = 0.4$, and real-speech prototypes are updated by EMA with momentum $\alpha = 0.88$.

Baselines & Protocol. We compare against: (i) Baseline (base corpora only); (ii) + Real TI (+ CommonVoice-ZH); (iii) + Synth TD (+ synthetic wake-word without domain adaptation); (iv) + Real TD (HI-MIA-trained cross-corpus reference); and (v) w/o Paired Real TI Anchors, in which the paired CommonVoice-ZH real utterances are removed while the synthesized wake-word data are retained. Because no same-speaker real targets or real prototypes are available in this configuration, the speaker-aware center-alignment loss is inactive. EER, minDCF ($p_{\text{tar}} = 0.01$), and low-FAR FRR are computed using cosine similarity without score normalization. Since evaluation follows a fixed-phrase TD-SV protocol, lexical mismatch detection is not considered in this work.

4.3. Evaluation Metrics

We use Equal Error Rate (EER) as the primary metric, complemented by minDCF ($p_{\text{tar}} = 0.01$) and the False Rejection Rate (FRR) at FAR = 1% and FAR = 0.1%, two low-FAR operating points particularly relevant to wake-word deployment. Trials are scored with cosine similarity over speaker embeddings without score normalization.

5. Results and Analysis

5.1. Main Results

SynthWake achieves 3.938% EER, a 5.5% relative improvement over the Baseline, and the lowest minDCF of 0.349. Its EER is also competitive with the cross-corpus + Real TD reference at 3.998%. Adding real TI data alone provides only a small gain (4.132% vs. 4.169%), whereas converting the same TI resource into wake-word speech improves EER to 4.007%. This confirms that synthesized phrase-specific speech provides more relevant supervision than simply adding further TI recordings.

The proposed alignment objectives further reduce EER from 4.007% to 3.938%. In contrast, removing the paired real TI anchors increases EER to 4.207%, above the Baseline. Without these anchors, neither current-batch real centroids nor real EMA prototypes are available, so speaker-aware alignment is inactive. Because this configuration also removes same-speaker overlap between the real adaptation and synthesized domains, it reflects the broader absence of paired real anchoring rather than the removal of one loss term alone.

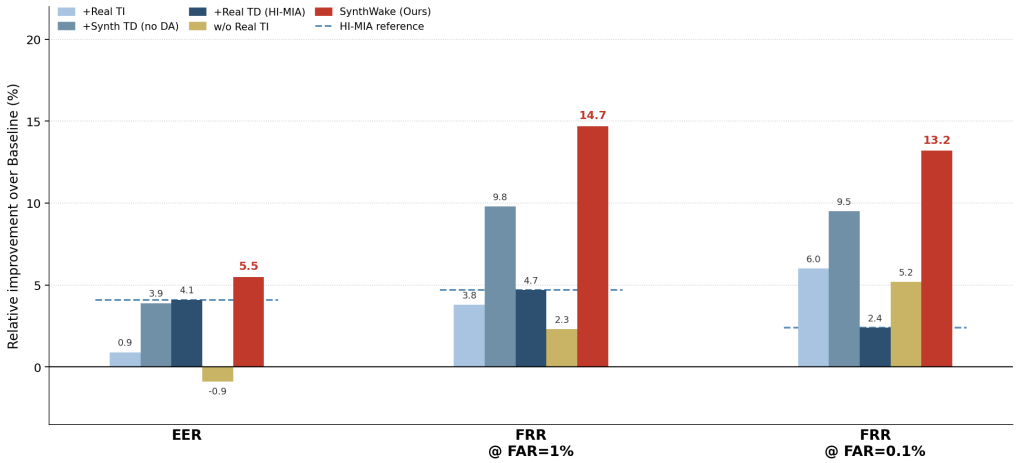


Figure 2: Relative reduction over the Baseline on SMIIP-TV. Dashed lines indicate the performance of the +Real TD (HI-MIA) reference. While the relative EER gains are moderate, larger improvements are observed at low-FAR operating points, where SynthWake achieves 14.7% and 13.2% relative FRR reductions at FAR = 1% and FAR = 0.1%, respectively. The negative EER reduction of *w/o Paired Real TI Anchors* suggests that paired real speech plays an important role in effective cross-domain alignment.

Table 3: Ablation study of SynthWake on SMIIP-TV. Mean ± std over 3 runs.

Configuration	EER (%)
Speaker CE loss only	4.007 ± 0.03
+ Cross-domain center alignment	3.951 ± 0.04
+ Domain adversarial training (GRL)	3.945 ± 0.02
+ Delayed synth. CE weighting	**3.938** ± **0.02**

Behavior at Low-FAR Operating Points. SynthWake reduces FRR by **14.7%** and **13.2%** relative to the Baseline at FAR = 1% and FAR = 0.1%, respectively, compared with 4.7% and 2.4% for + Real TD. The larger gains at strict operating points may be related to improved intra-speaker compactness and the additional acoustic variation provided by diverse TI prompts, although a detailed score-distribution analysis is left for future work.

5.2. *Ablation Study*

Table 3 presents the ablation results, with components added cumulatively from top to bottom.

Effect of Center Alignment. Adding cross-domain center alignment reduces EER from 4.007% to 3.951% (1.4% relative), showing that explicit centroid matching is the most impactful component for aligning real and LLM-synthesized embeddings. While GRL achieves global domain confusion, it does not guarantee that individual speaker clusters are well-aligned across domains; the per-speaker center alignment directly addresses this by pulling synthetic embeddings toward their real counterparts, which is necessary for the speaker discrimination task.

Effect of Domain Adaptation.. Domain adversarial training further reduces EER to 3.945%, providing a modest but consistent improvement. The smaller gain compared to center alignment (0.006 vs. 0.056 absolute EER points) suggests that per-speaker alignment is more effective than distribution-level confusion—likely because the verification decision is made at the speaker level rather than the domain level.

Effect of Delayed Synth. CE Weighting. Suppressing the synthesized-sample classification loss during the first epoch yields a marginal EER improvement (3.945% → 3.938%). Its primary benefit is training stability rather than final accuracy: activating synthesized-sample classification from the beginning resulted in less stable early optimization while the encoder was still forming its initial representation space. Enabling the synthesized-sample classification weight from the second epoch mitigates this behavior without sacrificing final performance.

Discussion. The ablation results show that cross-domain center alignment is the main source of improvement. This indicates that preserving speaker-level correspondence between real and synthesized speech is more important than enforcing global domain invariance alone. The additional gain from GRL is small but consistent, suggesting that domain confusion is useful only when combined with speaker-aware alignment. Delayed synthetic CE weighting contributes marginally to final EER, but it improves early-stage optimization stability.

5.3. Analysis and Discussion

Why synthesis from TI prompts performs competitively. One possible explanation is that TI prompts expose the TTS model to a broader range of speaker-specific acoustic and prosodic characteristics than short wake-word recordings. This may help preserve speaker identity during wake-word synthesis. In addition, using different TI prompts naturally introduces speaking-style variation into the synthesized wake-word corpus, increasing within-speaker diversity in a task-relevant form.

The paired training strategy may also contribute to the gain. Since real and synthesized utterances share the same speaker label, the encoder is encouraged to learn representations that are less sensitive to synthesis-related artifacts while preserving speaker identity. This provides a form of cross-domain consistency regularization that is not available when training only on real TD recordings.

Importance of paired real data. The degradation observed in the *w/o Paired Real TI Anchors* configuration highlights the importance of retaining same-speaker real speech during synthetic-data training. Without paired real utterances, neither current-batch real centroids nor real EMA prototypes can be constructed, and the speaker-aware alignment loss becomes inactive. Moreover, the real and synthesized adaptation domains no longer share speaker identities, causing the domain discriminator to capture both synthesis-related differences and differences in speaker composition. The model must consequently rely on speaker classification and marginal domain-adversarial learning without explicit speaker-wise real–synthetic correspondence.

The *w/o Paired Real TI Anchors* system still yields modest FRR reductions at low-FAR operating points, suggesting that synthetic TD data can provide useful phrase-specific supervision even without paired real anchors. However, because this configuration simultaneously changes the training data, speaker overlap, and availability of center alignment, its degradation should be interpreted as evidence for the overall importance of paired real anchoring rather than as an isolated estimate of the contribution of the alignment loss.

Failure Cases and Practical Constraints. We also explored two alternative synthesis strategies that did not improve performance. The first was to enlarge the speaker pool by sampling *novel synthetic identities* from IndexTTS and adding them to the classifier as additional classes. This performed below the Baseline, likely because these synthetic-only identities had no paired real counterparts and therefore could not benefit from speaker-level cross-domain alignment.

The second strategy replaced the wake-word synthesis target with *TI sentences*, treating IndexTTS as a generic text-independent augmenter. Its gains were comparable to simply adding more real TI data and were much smaller than those obtained by synthesizing the actual wake word. This suggests that the synthesized utterances need to match the lexical content of the deployed phrase. Combining synthetic identities with TI targets produced the worst result, indicating that both real speaker anchors and task-relevant synthesis targets are important for the current SynthWake design.

6. Conclusion

We proposed SynthWake, a data-efficient framework for wake-word verification. SynthWake uses zero-shot/LLM-based synthesis to convert TI recordings of real speakers into TD wake-word training data, and combines domain adversarial training with cross-domain centroid alignment to bridge the synthetic–real gap. The degradation of the *w/o Paired Real TI Anchors* system further shows that zero-shot TTS should not be treated as a drop-in replacement for real speaker data: real TI utterances are essential anchors for reliable speaker-level alignment. On SMIIP-TV, SynthWake attains 3.938% EER (5.5% relative improvement over the Baseline) and a 14.7% relative FRR reduction at FAR = 1%, achieving performance comparable to a real-TD cross-corpus reference while substantially reducing target-phrase data collection. The improvement is more pronounced under low-FAR operating conditions, suggesting that the proposed strategy may provide benefits beyond those reflected by the EER operating point alone.

Limitations. Validation is limited to Mandarin and a single wake word; multilingual and multi-phrase extensions, as well as tighter integration of synthesis-quality estimators into training, remain future work.

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